



Unit 1(Number system)

Computer Organization and Architecture (SRM Institute of Science and Technology)



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* NUMBER SYSTEMS *

* Number system is a way to represent numbers.

* The modern computers communicate and operate with binary numbers which use only digits 0 and 1.

* To represent long strings of bits in the addresses, instructions and data in digital systems, Octal and Hexadecimal number systems are used.

* The four number systems used in digital systems are

1. Decimal Number system.
2. Binary Number system.
3. Octal Number system.
4. Hexadecimal Number system.

* In any number system, the digits of a number has two parts. These two parts are integer and fractional part set apart by a radix point (.)

* In general, a number with a decimal point (.) is represented by a series of coefficients:

$$N = \underbrace{a_{n-1} a_{n-2} a_{n-3} \dots a_0}_{\text{Integer part}} \cdot \underbrace{a_{-1} a_{-2} \dots a_{-m}}_{\text{Fractional part}}$$

Here, a_{n-1} is Most Significant Bit (MSB) & a_{-m} is Least Significant Bit (LSB)

Radix/Base :

* The Base / radix of a number system is defined as the number of unique digits used to represent a number (including 0).

* In any number system, a number expressed in a base-r system has coefficients multiplied by the powers of 'r'.

$$a_n \cdot r^n + a_{n-1} \cdot r^{n-1} + \dots + a_2 \cdot r^2 + a_1 \cdot r^1 + a_0 \cdot r^0 + a_{-1} \cdot r^{-1} + a_{-2} \cdot r^{-2} + a_{-m} \cdot r^{-m}$$

* To distinguish numbers of different bases, we enclose the coefficients in parentheses and write the base value.

Decimal Number System	Binary Number System	Octal Number System	Hexadecimal Number System.
Base/Radix (r) = 10	Base/Radix (r) = 2	Base/Radix (r) = 8	Base/Radix (r) = 16
Range: 0 to 9	Range: 0 to 1	Range: 0 to 7	Range: 0 to F

NUMBER BASE CONVERSIONS.

1. Any Radix to Decimal:

* Conversion of any base to decimal is done by expanding the number in a power series by multiplying each digit in the number by its positional weight and adding all the terms.

2. Decimal to Any Radix:

* Convert integer part by successive Division Method.

* Convert fractional part by successive Multiplication Method.

3. Binary to Octal:

* Group of 3 bits replaced by its decimal equivalent.

4. Binary to Hexadecimal:

* Group of 4 bits replaced by its decimal equivalent.

5. Octal to Binary:

* Write 3-bit binary number ⁽⁴²¹⁾ equivalent to each octal digit and remove any leading and trailing zeros.

6. Hexadecimal to Binary:

* Write 4-bit binary number (8421) equivalent to each Hexadecimal digit and remove any leading and trailing zeros.

7. Octal to Hexadecimal:

* Octal $\rightarrow$ Binary $\rightarrow$ Hexadecimal

8. Hexadecimal to Octal:

* Hexadecimal $\rightarrow$ Binary $\rightarrow$ Octal.

9. Any Radix to Any Radix:

* Any Radix $\rightarrow$ Decimal $\rightarrow$ Another Radix.

① ANY RADIX TO DECIMAL NUMBER SYSTEM:

a) Binary to Decimal

$$(1101.101)_2 = (?)_{10}$$

Solution:

$$(1101.101)_2$$

$$\begin{aligned} 1 \times 2^{-3} &= 0.125 \\ 0 \times 2^{-2} &= 0.25 \\ 1 \times 2^{-1} &= 0.5 \\ 1 \times 2^0 &= 1 \\ 0 \times 2^1 &= 0 \\ 1 \times 2^2 &= 4 \\ 1 \times 2^3 &= 8 \end{aligned}$$

$$13.875$$

$$\therefore (1101.101)_2 = (13.875)_{10}$$

b) Octal to Decimal

$$(475.25)_8 = (?)_{10}$$

Solution:

$$(475.25)_8$$

$$\begin{aligned} 5 \times 8^{-2} &= 0.0781 \\ 2 \times 8^{-1} &= 0.25 \\ 5 \times 8^0 &= 5 \\ 7 \times 8^1 &= 56 \\ 4 \times 8^2 &= 256 \end{aligned}$$

$$317.3281$$

$$\therefore (475.25)_8 = (317.3281)_{10}$$

c) Hexadecimal to Decimal

$$(9B2.1A)_{16} = (?)_{10}$$

Solution:

$$(9B2.1A)_{16}$$

$$\begin{aligned} 10 \times 16^{-2} &= 0.0391 \\ 1 \times 16^{-1} &= 0.0625 \\ 2 \times 16^0 &= 2 \\ 11 \times 16^1 &= 176 \\ 9 \times 16^2 &= 2304 \end{aligned}$$

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d) $(3102.12)_4 = (?)_{10}$

Solution:

$$(3102.12)_4$$

$$\begin{aligned} 2 \times 4^{-2} &= 0.125 \\ 1 \times 4^{-1} &= 0.25 \\ 2 \times 4^0 &= 2 \\ 0 \times 4^1 &= 0 \\ 1 \times 4^2 &= 16 \\ 3 \times 4^3 &= 192 \end{aligned}$$

$$210.375$$

$$\therefore (3102.12)_4 = (210.375)_{10}$$

e) $(4021.2)_5 = (?)_{10}$

Solution:

$$(4021.2)_5$$

$$\begin{aligned} 2 \times 5^{-1} &= 0.4 \\ 1 \times 5^0 &= 1 \\ 2 \times 5^1 &= 10 \\ 0 \times 5^2 &= 0 \\ 4 \times 5^3 &= 500 \end{aligned}$$

$$511.4$$

$$\therefore (4021.2)_5 = (511.4)_{10}$$

② DECIMAL TO ANY RADIX

Example: $(5679.329)_{10} =$ a) $(?)_2$ b) $(?)_8$ c) $(?)_{16}$ d) $(?)_9$

Integer part: 5679, Fractional part: 329

↓
Successive Division
Method

↓
Successive Multiplication
Method.

Solution:

a) $(5679.329)_{10} = (?)_2$

* Applying successive division on Integer part.

$$\begin{array}{r}
 2 \overline{) 5679} \\
 2 \overline{) 2839} \quad -1 \\
 2 \overline{) 1419} \quad -1 \\
 2 \overline{) 709} \quad -1 \\
 2 \overline{) 354} \quad -1 \\
 2 \overline{) 177} \quad -0 \\
 2 \overline{) 88} \quad -1 \\
 2 \overline{) 44} \quad -0 \\
 2 \overline{) 22} \quad -0 \\
 2 \overline{) 11} \quad -0 \\
 2 \overline{) 5} \quad -1 \\
 2 \overline{) 2} \quad -1 \\
 1 \quad -0
 \end{array}$$

$$(5679)_{10} = (1011000101111)_2$$

* Applying successive multiplication on Fractional part.

$$\begin{array}{l}
 0.329 \times 2 = 0.648 \\
 0.648 \times 2 = 1.296 \\
 0.296 \times 2 = 0.592 \\
 0.592 \times 2 = 1.084
 \end{array}$$

$$(0.329)_{10} = (0.0101)_2$$

$$\therefore (5679.329)_{10} = (1011000101111.0101)_2$$

b) $(5689.329)_{10} = (?)_8$

* Applying successive division on Integer part

$$\begin{array}{r}
 8 \overline{) 5689} \\
 8 \overline{) 711} \quad -1 \\
 8 \overline{) 88} \quad -7 \\
 8 \overline{) 11} \quad -0 \\
 1 \quad -3
 \end{array}$$

$$(5689)_{10} = (13071)_8$$

* Applying successive multiplication on Fractional part.

$$\begin{array}{l}
 0.329 \times 8 = 2.632 \\
 0.632 \times 8 = 5.056 \\
 0.056 \times 8 = 0.448 \\
 0.448 \times 8 = 3.584
 \end{array}$$

$$(0.329)_{10} = (0.2503)_8$$

$$\therefore (5689.329)_{10} = (13071.2503)_8$$

c) $(5689.329)_{10} = (?)_{16}$

* Applying successive division on Integer part.

$$\begin{array}{r}
 16 \overline{) 5689} \\
 16 \overline{) 355} \quad -9 \\
 16 \overline{) 22} \quad -3 \\
 1 \quad -6
 \end{array}$$

$$(5689)_{10} = (1639)_{16}$$

* Applying successive multiplication on Fractional part.

$$\begin{array}{l}
 0.329 \times 16 = 5.264 \\
 0.264 \times 16 = 4.224 \\
 0.224 \times 16 = 3.584 \\
 0.584 \times 16 = 9.344
 \end{array}$$

$$(0.329)_{10} = (0.5439)_{16}$$

$$d) (5689.329)_{10} = (?)_9$$

* Applying successive division on Integer part

$$\begin{array}{r} 9 \overline{) 5689} \\ 9 \overline{) 632} - 1 \uparrow \\ 9 \overline{) 70} - 2 \\ 7 - 7 \end{array}$$

$$(5689)_{10} = (7721)_9$$

$$\therefore (5689.329)_{10} = (7721.2857)_9$$

* Applying successive multiplication on Fractional part.

$$\begin{array}{l} 0.329 \times 9 = 2.961 \\ 0.961 \times 9 = 8.649 \\ 0.649 \times 9 = 5.841 \\ 0.841 \times 9 = 7.569 \downarrow \end{array}$$

$$(0.329)_{10} = (2857)_9.$$

③ BINARY TO OCTAL CONVERSION.

Step 1:

- * For Integer part, make group of 3 bits starting from LSB
- * For Fractional part, make group of 3 bits starting from MSB

Step 2:

- * Add 0's at the ends if required and convert 3 bits into decimal equivalent.

Example:

$$1) (1001011.1011)_2 = (?)_8$$

Solution:

$$\begin{array}{ccccccc} 00 & 1 & 001 & 011 & 101 & 100 \\ & & \leftarrow & & \rightarrow & \\ & 1 & 1 & 3 & 5 & 4 \end{array}$$

$$\therefore (1001011.1011)_2 = (113.54)_8$$

$$2) (1101.11)_2 = (?)_8$$

Solution:

$$\begin{array}{cccc} 00 & 1 & 101 & 110 \\ \leftarrow & & \rightarrow & \\ & 1 & 5 & 6 \end{array}$$

$$\therefore (1101.11)_2 = (15.6)_8$$

④ BINARY TO HEXADECIMAL CONVERSION

Step 1:

- * For Integer part, make group of 4-bits starting from LSB
- * For Fractional part, make group of 4-bits starting from MSB

Step 2:

- * Add 0's at the ends if required and convert 4 bits into decimal equivalent.

Example

$$1) (110100111.111101)_2 = (?)_{16}$$

Solution:

$$\begin{array}{c|c|c|c|c} \underline{00} & 11 & 01 & 00 & 111 \\ \hline & 1 & A & 7 & . & F & 4 \end{array}$$

$$\therefore (110100111.111101)_2 = (1A7.F4)_{16}$$

$$2) (1101101110.10011010)_2 = (?)_{16}$$

Solution:

$$\begin{array}{c|c|c|c|c} \underline{00} & 11 & 011 & 011 & 10 \\ \hline & 3 & 6 & E & . & 9 & A \end{array}$$

$$\therefore (1101101110.10011010)_2 = (36E.9A)_{16}$$

(5) OCTAL TO BINARY CONVERSION

Step 1: Write the 3 bit binary equivalent to each octal digit.

Step 2: Remove any trailing and leading zeros.

Example

$$1) (125.62)_8 = (?)_2$$

Solution

$$\begin{array}{c|c|c|c|c} 1 & 2 & 5 & . & 6 & 2 \\ \hline 001 & 010 & 101 & & 110 & 010 \end{array}$$

$$= (1010101.110010)_2$$

$$\therefore (125.62)_8 = (1010101.110010)_2$$

$$2) (634.27)_8 = (?)_2$$

$$\begin{array}{c|c|c|c|c} 6 & 3 & 4 & . & 2 & 7 \\ \hline 110 & 011 & 100 & & 010 & 111 \end{array}$$

$$= (110011100.010111)_2$$

$$\therefore (634.27)_8 = (110011100.010111)_2$$

⑥ HEXADECIMAL TO BINARY CONVERSION

Step 1: Write the 4-bit binary equivalent to each hexadecimal digit.

Step 2: Remove any trailing and leading zeros.

Example

$$1) (8A9.B4)_{16} = (?)_2$$

Solution

$$\begin{array}{c|c|c|c|c} 8 & A & 9 & . & B & 4 \\ \hline 1000 & 1010 & 1001 & & 1011 & 0100 \end{array}$$

$$= (100010101001.101101)_2$$

$$\therefore (8A9.B4)_{16}$$

$$= (100010101001.101101)_2$$

$$2) (1FF3.00A)_{16} = (?)_2$$

Solution

$$\begin{array}{c|c|c|c|c|c|c} 1 & F & F & 3 & . & 0 & 0 & A \\ \hline 0001 & 1111 & 1111 & 0011 & & 0000 & 0000 & 1010 \end{array}$$

$$= (111111110011.00000000101)_2$$

$$\therefore (1FF3.00A)_{16} = (111111110011.00000000101)_2$$

⑦ OCTAL TO HEXADECIMAL CONVERSION

Step 1: Convert each octal digit to its 3 digit binary equivalent.

Step 2: Now convert this binary number to hexadecimal by making group of 4-bits.

Step 3: Add zeros at the ends if required & write its hexadecimal equivalent

Example

$$1) (615.25)_8 = (?)_{16}$$

Solution:

$$\text{Octal} \rightarrow 615.25$$

$$\text{Binary} \rightarrow 110001101.010101$$

Converting Binary to Hexadecimal

$$\begin{array}{c|c|c|c} 0001 & 1000 & 1101 & . & 0101 & 0100 \\ \hline 1 & 8 & D & . & 5 & 4 \end{array}$$

$$2) (167.32)_8 = (?)_{16}$$

Solution:

$$\text{Octal} \rightarrow 167.32$$

$$\text{Binary equivalent} \rightarrow 001110111.011010$$

Converting Binary to Hexadecimal

$$\begin{array}{c|c|c|c} 0000 & 0111 & 0111 & . & 0110 & 1000 \\ \hline 0 & 7 & 7 & . & 6 & 8 \end{array}$$

$$\therefore (167.32)_8 = (77.68)_{16}$$

⑧ HEXADECIMAL TO OCTAL CONVERSION

Step 1: Convert each hexadecimal digit into its 4-bit binary equivalent

Step 2: Convert this binary number into octal by making a group of 3 bits.

Step 3: Add trailing zeros at the ends if necessary and write its octal equivalent.

Example

① $(BC66.AF)_{16} = (?)_8$

Solution:

Hexadecimal $\rightarrow BC66.AF$

Binary equivalent $\rightarrow 1011110001100110.10101111$

Converting Binary to Octal:

001	011	110	001	100	110	.	101	011	110
1	3	6	1	4	6	.	5	3	6

$\therefore (BC66.AF)_{16} = (136146.536)_8$

② $(FF3.AB)_{16} = (?)_8$

Solution:

Hexadecimal $\rightarrow FF3.AB$

Binary equivalent $\rightarrow 111111110011.10101011$

Converting Binary to Octal:

111	111	110	011	.	101	010	110
7	7	6	3	.	5	2	6

$\therefore (FF3.AB)_{16} = (7763.526)_8$

⑨ ANY RADIX TO ANY OTHER RADIX

Step 1 : Convert the number in any radix to Decimal

Step 2 : Convert from Decimal to the other Radix.

Example

1) Convert $(432.34)_5$ to $(?)_6$

Step 1: Any Radix to Decimal

$$\begin{array}{cccccc}
 5^2 & 5^1 & 5^0 & 5^{-1} & 5^{-2} & \\
 4 & 3 & 2 & 3 & 4 & \\
 & & & & \rightarrow 4 \times 5^{-2} = 0.16 \\
 & & & \rightarrow 3 \times 5^{-1} = 0.6 \\
 & & \rightarrow 2 \times 5^0 = 2 \\
 & \rightarrow 3 \times 5^1 = 15 \\
 \rightarrow 4 \times 5^2 = 100
 \end{array}$$

$$\therefore (432.34)_5 = (117.76)_{10}$$

$$117.76$$

Step 2: Decimal to Base 6

$$(117.76)_{10} = (?)_6$$

* Successive division on Integer part

$$\begin{array}{r}
 6 \overline{) 117} \\
 6 \overline{) 19} - 3 \\
 \quad 3 - 1
 \end{array}
 \quad (117)_{10} = (313)_6$$

* Successive Multiplication on Fractional part

$$\begin{array}{rcl}
 0.76 \times 6 & = & 4.56 \\
 0.56 \times 6 & = & 3.36 \\
 0.36 \times 6 & = & 2.16
 \end{array}
 \downarrow$$

$$(0.76)_{10} = (0.432)_6$$

$$(117.76)_{10} = (313.432)_6$$

$$\therefore (432.34)_5 = (117.76)_{10} = (313.432)_6$$

2) Convert $(213.03)_4$ to $(?)_7$

Step 1: Any Radix to Decimal

$$\begin{array}{cccccc}
 4^2 & 4^1 & 4^0 & 4^{-1} & 4^{-2} & \\
 2 & 1 & 3 & 0 & 3 & \\
 & & & & \rightarrow 3 \times 4^{-2} = 0.1875 \\
 & & & \rightarrow 0 \times 4^{-1} = 0 \\
 & & \rightarrow 3 \times 4^0 = 3 \\
 & \rightarrow 1 \times 4^1 = 4 \\
 \rightarrow 2 \times 4^2 = 32
 \end{array}$$

$$39.1875$$

$$\therefore (213.03)_4 = (39.1875)_{10}$$

Step 2: Decimal to Base 7

$$(39.1875)_{10} = (?)_7$$

* Successive division on Integer part

$$\begin{array}{r}
 7 \overline{) 39} \\
 \quad 5 - 4
 \end{array}
 \uparrow$$

$$(39)_{10} = (54)_7$$

* Successive Multiplication on Fractional part

$$\begin{array}{rcl}
 0.1875 \times 7 & = & 1.3125 \\
 0.3125 \times 7 & = & 2.1875 \\
 0.1875 \times 7 & = & 1.3125
 \end{array}
 \downarrow$$

$$(0.1875)_{10} = (0.121)_7$$

$$(39.1875)_{10} = (54.121)_7$$

$$(213.03)_4 = (39.1875)_{10} = (54.121)_7$$

* ARITHMETIC OPERATIONS *

* BINARY ADDITION *

Rules for Binary Addition

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

① Add $(1010)_2$ and $(111)_2$

Solution

$$\begin{array}{r} 1010 \\ + 111 \\ \hline 10001 \end{array}$$

$$\therefore (1010)_2 + (111)_2 = (10001)_2$$

② $1101 \cdot 101 + 111 \cdot 011 = ?$

Solution

$$\begin{array}{r} 1101 \cdot 101 \\ + 111 \cdot 011 \\ \hline 10101 \cdot 000 \end{array}$$

$$\therefore (1101 \cdot 101)_2 + (111 \cdot 011)_2 = (10101 \cdot 000)_2$$

③ Add $(37)_{10}$ and $(56)_{10}$ in binary

Solution:

Convert decimal numbers to Binary

$$\begin{array}{r} 2 \overline{) 37} \\ 2 \overline{) 18-1} \\ 2 \overline{) 9-0} \\ 2 \overline{) 4-1} \\ 2 \overline{) 2-0} \\ 1-0 \end{array}$$

$$\begin{array}{r} 2 \overline{) 56} \\ 2 \overline{) 28-0} \\ 2 \overline{) 14-0} \\ 2 \overline{) 7-0} \\ 2 \overline{) 3-1} \\ 1-1 \end{array}$$

$$(37)_{10} = (100101)_2$$

$$(56)_{10} = (111000)_2$$

$$(37)_{10} \quad 100101$$

$$(56)_{10} \quad 111000$$

$$\boxed{1011101} \text{ Ans.}$$

To verify

$$37 + 56 = 93$$

Result of Binary Addition = 1011101

$$1 \times 2^6 + 0 \times 2^5 + 1 \times 2^4 + 1 \times 2^3 + 1 \times 2^2 + 0 \times 2^1 + 1 \times 2^0$$

$$= 64 + 0 + 16 + 8 + 4 + 0 + 1$$

$$= 93$$

$$\therefore (37)_{10} + (56)_{10} = (93)_{10}$$

④ Add $1010 \cdot 11_2 + 1101 \cdot 10_2 +$

$$1001 \cdot 11_2 + 1111 \cdot 11_2$$

Solution:

$$\begin{array}{r} 1010 \cdot 11 \\ + 1101 \cdot 10 \\ + 1001 \cdot 11 \\ + 1111 \cdot 11 \\ \hline \end{array}$$

$$(110001 \cdot 11)_2 \text{ Ans.}$$

$$\begin{array}{r} 101011 \\ + 110110 \\ + 100111 \\ + 111111 \\ \hline \end{array}$$

* BINARY MULTIPLICATION *

Rules for Binary Multiplication

A	B	A · B
0	0	0
0	1	0
1	0	0
1	1	1

- ① Multiply $(1101)_2$ with $(110)_2$ ② Multiply $1011 \cdot 101$ with $101 \cdot 01$

$$\begin{array}{r}
 1101 \\
 \times 110 \\
 \hline
 0000 \\
 1101 \\
 1101 \\
 \hline
 1001110
 \end{array}$$

$$\therefore (1101)_2 \times (110)_2 = (1001110)_2$$

$$\begin{array}{r}
 1011 \cdot 101 \\
 \times 101 \cdot 01 \\
 \hline
 1011101 \\
 , 0000000 \\
 , 1011101 \\
 0000000 \\
 \hline
 1011101 \\
 \hline
 11110100001
 \end{array}$$

$$\therefore (1011 \cdot 101)_2 \times (101 \cdot 01)_2 = (111101 \cdot 00001)_2$$

* BINARY DIVISION *

Rules for Binary Division

$$\begin{array}{l}
 0 \div 1 = 0 \\
 1 \div 1 = 1
 \end{array}$$

- ① Divide $(11011011)_2$ by $(110)_2$

$$\begin{array}{r}
 100100 \cdot 1 \\
 110 \overline{) 11011011} \\
 \underline{110} \\
 0110 \\
 \underline{110} \\
 0110 \\
 \underline{110} \\
 0 \\
 \hline
 0
 \end{array}$$

$$\therefore (11011011) \div (110) =$$

$$(100100 \cdot 1)_2$$

- ② $(110101 \cdot 11)_2 \div (101)_2$

$$\begin{array}{r}
 1010 \cdot 11 \\
 101 \overline{) 110101 \cdot 11} \\
 \underline{101} \\
 110 \\
 \underline{101} \\
 11 \\
 \underline{101} \\
 101 \\
 \underline{101} \\
 0 \\
 \hline
 0
 \end{array}$$

$$\begin{array}{l}
 0-0=0 \\
 0-1=1 \text{ borrow} \\
 1-0=1 \\
 1-1=0
 \end{array}$$

$$\therefore (110101 \cdot 11)_2 \div (101)_2 = (1010 \cdot 11)_2$$

* BINARY SUBTRACTION *

Rules for Binary subtraction

A	B	Diff	Borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$10 - 1 = 1$$

1) $(11101100)_2 - (00110010)_2$

$$\begin{array}{r} 11101100 \\ (-) 00110010 \\ \hline 10111010 \end{array} \text{Ans}$$

$$\therefore (11101100)_2 - (00110010)_2 = (10111010)_2$$

2) $(1110)_2 - (1101)_2$

$$\begin{array}{r} 1110 \\ (-) 1101 \\ \hline 0001 \end{array} \text{Ans}$$

$$\therefore (1110)_2 - (1101)_2 = (0001)_2$$

3) $(10001.01)_2 - (1110.10)_2$

$$\begin{array}{r} 10001.01 \\ (-) 1110.10 \\ \hline 00010.11 \end{array} \text{Ans}$$

$$\therefore (10001.01)_2 - (1110.10)_2 = (00010.11)_2$$

4) $(1001.11)_2 - (1000.11)_2$

$$\begin{array}{r} 1001.11 \\ (-) 1000.11 \\ \hline 0001.00 \end{array} \text{Ans}$$

$$\therefore (1001.11)_2 - (1000.11)_2 = (0001.00)_2$$

* COMPLEMENTS OF NUMBERS *

* In digital computers, complements are used to simplify the subtraction operation.

* There are two types of complements for each base-r system.

1. The Radex complement (r 's complement)

2. Diminished radex complement ($r-1$'s complement).

* Given a number N in base ' r ' having n digits.

$$\Rightarrow r-1 \text{ complement} = (r^n - 1) - N$$

$$\Rightarrow r \text{'s complement} = r^n - N. \text{ (ie) } (r-1 \text{ complement} + 1)$$

* SUBTRACTION WITH COMPLEMENTS *

Rules for subtraction using Complements:

The subtraction of two n -digit numbers $M-N$ in base ' r ' can be done as follows:

- 1) Add the minuend M to the r 's complement / $(r-1)$'s complement of the subtrahend N .
- 2) If $M \geq N$, the sum will produce an end carry.
 - In case of 1's and 9's complement, add end carry to sum which gives the final result. (i.e. $r-1$ complement)
 - In case of 2's and 10's complement, discard the end carry. (i.e. r complement)
- 3) If $M < N$, the sum does not produce an end carry. The result is the r 's complement of $(N-M)$.
 - To obtain the answer in a familiar form, take the r 's complement of the sum and place a negative sign in front.

① Using 10's complement, subtract $72532 - 3250$.

Solution:

Given $M = 72532$, $N = 03250$

Find 10's complement of N :

10's complement = 9's complement + 1

$$\begin{array}{r} 9's \text{ complement} \Rightarrow 99999 \\ \phantom{9's \text{ complement} \Rightarrow} 03250 \\ \hline 96749 \end{array}$$

$$\begin{array}{r} 10's \text{ complement} \Rightarrow 9 \\ \phantom{10's \text{ complement} \Rightarrow} 96749 \\ \hline 96750 \end{array}$$

$$M = 72532$$

$$\begin{array}{r} 10's \text{ complement} \stackrel{(*)}{=} 96750 \\ \text{of } N \quad \underline{\underline{069282}} \end{array}$$

End carry is produced. In case of r 's complement (10's complement) the carry can be discarded.

$$\therefore 72532 - 3250 = 69282$$

② Using 10's complement, subtract $3250 - 72532$.

Solution:

Given $M = 03250$, $N = 72532$

Find 10's complement of N :

10's complement = 9's complement + 1

$$\begin{array}{r} 9's \text{ complement} \Rightarrow 99999 \\ \phantom{9's \text{ complement} \Rightarrow} - 72532 \\ \hline 27467 \end{array}$$

$$\begin{array}{r} 10's \text{ complement} \Rightarrow 6 \\ \phantom{10's \text{ complement} \Rightarrow} 27467 \\ \hline 27468 \end{array}$$

$$M = 03250$$

$$\begin{array}{r} 10's \text{ comp} \stackrel{(*)}{=} 27468 \\ \text{of } N \quad \underline{\underline{30718}} \end{array}$$

There is no end carry.

$\therefore$ The answer is - (10's complement of 30718)

$$\begin{array}{r} 10's \text{ complement of } 30718 = 99999 \\ \phantom{10's \text{ complement of } 30718} 30718 \\ \hline 69281 \\ \phantom{10's \text{ complement of } 30718} + 1 \\ \hline 69282 \end{array}$$

Finding Complements of Base-r numbers:

- ① Consider the Binary Number $N = (101100)_2$. Find r 's complement and $(r-1)$'s complement.

Solution:

Given a Binary number. Radix $r = 2$.

1's complement: $(r-1)$ complement

* It is found by subtracting each digit of N from 1

$$\begin{array}{r} 111111 \\ (-) 101100 \\ \hline 010011 \end{array} \rightarrow 1's \text{ complement}$$

- (or) 1's complement of a Binary Number N is formed by changing 1's to 0's and 0's to 1's.

2's complement: (r) complement

2's complement = 1's complement + 1

$$\begin{array}{r} 010011 \\ + 1 \\ \hline 010100 \end{array} \rightarrow 2's \text{ complement of } N.$$

- ② Consider the Decimal Number $N = (546700)_{10}$. Find its r 's complement and $(r-1)$'s complement.

Solution:

Given a Decimal Number. Radix $r = 10$.

9's complement: $(r-1)$ complement

* It is found by subtracting each digit of N from $(r-1)$ (i.e.) 9

$$\begin{array}{r} 999999 \\ (-) 546700 \\ \hline 453299 \end{array} \rightarrow 9's \text{ complement}$$

10's complement: (r) complement

10's complement = 9's complement + 1

$$453299$$

- ③ Consider the Octal Number $N = (756214)_8$. Find its r 's complement and $(r-1)$'s complement.

Solution:

Given the Octal Number. Radix $r = 8$.

7's complement: $(r-1)$ complement

* It is found by subtracting each digit of N from 7 ($r-1$)

$$\begin{array}{r} 777777 \\ (-) 756214 \\ \hline 021563 \end{array} \rightarrow 7's \text{ complement}$$

8's complement: (r) complement

8's complement = 7's complement + 1

$$\begin{array}{r} 021563 \\ + 1 \\ \hline 021564 \end{array} \rightarrow 8's \text{ complement.}$$

- ④ Consider a Hexadecimal Number $N = (5AF624)_{16}$. Find its r 's complement and $(r-1)$'s complement.

Solution:

Given Hexadecimal Number. Radix $r = 16$

15's complement: $(r-1)$ complement

* It is found by subtracting each digit of N from 15 (F)

$$\begin{array}{r} FFF FFF \quad (15) \\ (-) 5AF624 \\ \hline A509DB \end{array} \rightarrow 15's \text{ complement}$$

16's complement: (r) 's complement

16's complement = 15's complement + 1

$$\begin{array}{r} A509DB \\ + 1 \\ \hline A509DC \end{array} \rightarrow 16's \text{ complement}$$

*** Similarly r 's complement and $(r-1)$'s complement can be found for

③ Given two binary numbers:
 $X = 1010100$ and $Y = 1000011$, Perform
 the subtraction a) $X - Y$ and b) $Y - X$
 using 2's complements.

Solution:

a) $X - Y$

Given $X = 1010100$, $Y = 1000011$

Find 2's complement of Y

2's complement = 1's complement + 1

1's complement of $Y = 0111100$

2's complement of $Y = 0111100$

$$\begin{array}{r} + 1 \\ 0111100 \\ \hline \end{array}$$

$X = 1010100$

2's complement of $Y = 0111101$

10010001

End carry is produced. In case of 1's complement (2's complement here), the carry can be discarded from the sum.

$$\therefore (1010100)_2 - (1000011)_2 = (0010001)_2$$

b) $Y - X$

Find 2's complement of X

2's complement = 1's complement + 1

1's complement of $X = 0101011$

2's complement of $X = 0101011$

$$\begin{array}{r} + 1 \\ 0101011 \\ \hline \end{array}$$

$Y = 1000011$

2's complement of $X = 0101100$ (+)

1101111

No End Carry. $Y - X = -(2's \text{ complement of } X)$

④ Given two binary Numbers
 $X = 1010100$ and $Y = 1000011$. Perform
 a) $X - Y$ and b) $Y - X$ using 1's complement

Solution:

a) $X - Y$

Given $X = 1010100$, $Y = 1000011$

Find 1's complement of Y

1's complement = 0111100

$X = 1010100$

1's comp of $Y = 0111100$

10010000

End carry is produced. In case of (1's) complement (1's complement here), the end carry is added to the sum to produce final result

Sum = 0010000

End carry = 1

$$\therefore X - Y = (0010001)_2 \text{ Ans}$$

b) $Y - X$

Find 1's complement of X

1's complement of $X = 0101011$

$Y = 1000011$

1's comp of $X = 0101011$ (+)

1101110

No End Carry. $Y - X = -(1's \text{ complement of } 1101110)$

$$\therefore Y - X = -(0010001)_2 \text{ Ans}$$

* SIGNED BINARY NUMBERS *

* The signed numbers has a symbol (+ or -) or a bit (0 or 1) indicating the sign.

* Positive number start with 0. Eg: +5 = $\overset{\text{sign bit}}{0}000\ 0101$

* Negative number start with 1. Eg: -5 = $1000\ 0101$

* The leftmost bit is the sign bit for signed Binary Numbers.

Examples:

① (+6) + (+13)

$$\begin{array}{r} +6 \quad \quad 0000\ 0110 \\ +13 \quad (+) \quad 0000\ 1101 \\ \hline +19 \quad \quad \underline{0001\ 0011} = (19)_{10} \end{array}$$

② (-6) + (+13)

Find 2's complement of -6

Binary of -6 = $1000\ 0110$

1's complement = $1111\ 1001$

2's complement = $\underline{1111\ 1010}$

$$\begin{array}{r} -6 \quad \quad 1111\ 1010 \\ +13 \quad 0000\ 1101 (+) \\ \hline +7 \quad \times \quad \underline{0000\ 0111} = (7)_{10} \end{array}$$

③ (+6) + (-13)

Find 2's complement of -13

Binary of -13 = $1000\ 1101$

1's complement = $1111\ 0010$

2's complement = $\underline{1111\ 0011}$

+6 $\quad 0000\ 0110$

-13 (+) $\quad 1111\ 0011$

④ (-6) + (-13)

Find 2's complement of -6

Binary of -6 = $1000\ 0110$

1's complement = $1111\ 1001$

2's complement = $\underline{1111\ 1010}$

Find 2's complement of -13

Binary of -13 = $1000\ 1101$

1's complement = $1111\ 0010$

2's complement = $\underline{1111\ 0011}$

$$\begin{array}{r} -6 \quad \quad 1111\ 1010 \\ -13 \quad (+) \quad 1111\ 0011 \\ \hline -19 \quad \quad \underline{1110\ 1101} = (-19) \end{array}$$

$\therefore (1111\ 1011)_2 = 2's \text{ complement of } +19$

Verify: $0001\ 0011 \Rightarrow \text{Binary of } 19$

1's complement of 19 = $1110\ 1100$

2's complement of 19 = $\underline{1110\ 1101}$

-19 = $\underline{1110\ 1101}$

$\therefore (-6) + (-13) = (1110\ 1101)_2$

* BINARY CODES *

* A Binary code represents a coding system using the binary digits 0 and 1 to represent a letter, digit or a character in an electronic device.

* "A n-bit binary code is a group of n-bits that assumes upto 2^n distinct combinations of 1's and 0's, with each combination representing one element of the set that is being coded."

* Binary codes are broadly classified into

1. Numeric codes

i) Weighted codes. (BCD 8421, 5421, 2421, 84-2-1)

ii) Non-weighted codes (Excess-3, Gray code)

2. Alphanumeric codes (ASCII)

3. Error detecting codes.

① Binary - Coded Decimal (BCD) code: (Weighted Code)

* Although the binary number system is the natural system for a computer, most of us are more accustomed to the decimal system.

* The difference is resolved by converting decimal numbers (0 to 9) to binary, perform all arithmetic calculations in binary and then convert the binary results back to decimal.

* BCD is a Weighted Numeric code. It is the most commonly used code.

* BCD is a code in which each digit of a decimal number is represented by a separate group of 4-bits.

Decimal Digit	BCD 8421	5421	2421	84-2-1
0	0000	0000	0000	0000
1	0001	1001	0001	0111
2	0010	0010	0010	0110
3	0011	0011	0011	0101
4	0100	0100	0100	0100
5	0101	1000	1011	1001
6	0110	1001	1000	1010
7	0111	1010	1001	1001
8	1000	1011	1101	1000
9	1001	1010	1110	1000

* BCD is a weighted code in which each bit position is assigned a weighting factor in such a way that each digit can be evaluated by adding the weights of all the 1's in the coded combination.

5421 code:

* The bit combination 0101, when weighted by 5421 code is

$$5 \times 0 + 4 \times 1 + 2 \times 0 + 1 \times 1 = 0 + 4 + 1 = 5 \text{ (decimal equivalent)}$$

* Some digits can be coded in two possible ways in the 5421 code.

Eg: Decimal 6 can be assigned to bit combination 1001 or 0110

2421 code:

* The bit combination 1101, when weighted by 2421 code is

$$= 2 \times 1 + 4 \times 1 + 2 \times 0 + 1 \times 0 = 2 + 4 = 6 \text{ (decimal equivalent)}$$

* Some digits can be coded in two possible ways in 2421 code.

Eg: Decimal 4 can be assigned to bit combination 0100 or 1010.

84-2-1 code:

* 84-2-1 code assigns both positive and negative weights to a decimal code.

* The bit combination 0110 is interpreted as decimal 2 in 84-2-1

$$8 \times 0 + 4 \times 1 + (-2) \times 1 + -1 \times 0 = 4 - 2 = 2.$$

8421 code: (commonly used BCD code)

* The decimal equivalent for 0110 in 8421 code is calculated as

$$8 \times 0 + 4 \times 1 + 2 \times 1 + 1 \times 0 = 4 + 2 = 6.$$

② Excess-3 Codes (Unweighted codes)

* unweighted code (X)

* Each coded combination is obtained as corresponding binary value + 3.

Decimal	Binary	Excess-3
0	0000	0011
1	0001	0100
2	0010	0101
3	0011	0110
4	0100	0111
5	0101	1000
6	0110	1001
	0111	1010

③ Gray code (Unweighted Code)

* In gray code, only one bit changes when going from one number to the next.

* It is convenient to use the gray code in some types of analog to digital converters.

Decimal	Gray code
0	0000
1	0001
2	0011
3	0010
4	0110
5	0111
6	0101
7	0100
8	1100
9	1101
10	1111
11	1110
12	1010
13	1011
14	1001
15	1000

④ ASCII Character Code:

* The alphanumeric codes represents alphabetic and numeric information as a series of 1's and 0's.

* The standard binary code for the alphanumeric characters is American Standard Code for Information Interchange (ASCII).

* ASCII is a 7-bit code and so 2^7 or 128 characters can be represented.

* The 7-bit code is used to represent 128 alphanumeric characters (i.e) 26 upper case letters, 26 lower case letters, 10 numerals (0-9), 32 printable characters and 34 control characters.

⑤ Error Detecting Code:

* When the digital information in the binary form is transmitted from one system to another, an error may occur.

* This means a signal corresponding to 0 may change to 1 or vice versa due to the presence of noise.

Parity Bit:

* The simplest technique for detecting error is to add an extra bit known as parity bit. It is like the even or odd number of 1's in a byte.

Example:

Consider the following ASCII values and their odd and even parity.

	With even parity	With odd parity
ASCII = 1000001	01000001	11000001
ASCII = 1010100	11010100	01010100

* The eight-bit characters that include parity bits are transmitted to their destination.

* The parity of each character is then checked at the receiving end.

* If the parity of the received character is not even, then at least one bit has changed in value during the transmission is detected.

* If an error is detected, the receiver may request for retransmission of the message by sending back ASCII NAK (-ve Ack)

* If no error is detected, the receiver sends back an ACK (acknowledgement) control character.

* BCD ADDITION *

* It is a numeric code in which each digit of a decimal number is represented by separate group of bits.

Example :

Binary value of 18 : $2^4 \quad 2^3 \quad 2^2 \quad 2^1 \quad 2^0$
10010

BCD Value of 18 : 0001 1000

Procedure for BCD Addition:

Step 1: Write BCD equivalent for every decimal digit and perform addition using ordinary Addition rules.

Step 2 : a) If sum ≤ 9 & No carry, the sum is in valid BCD form

b) If sum ≥ 10 & No carry, the result is invalid BCD digit. So, add 6 (0110) to get valid BCD digit and a BCD carry.

occurs when
sum ≥ 16 $\leftarrow$ c) If sum ≤ 9 , with carry, the result is invalid and needs correction because of the carry. So, add 6 (0110) to get the valid BCD sum and a BCD carry.

Examples for BCD Addition:

Pg. No: 23

① $4 + 5$

Solution:

$$\begin{array}{r} 4 \\ + 5 \\ \hline 9 \end{array} \quad \begin{array}{r} 0100 \\ + 0101 \\ \hline 1001 \end{array} \quad \begin{array}{l} \text{Ans BCD sum} \\ (9) \end{array}$$

② $4 + 8$

Solution:

$$\begin{array}{r} 4 \\ + 8 \\ \hline 12 \end{array} \quad \begin{array}{r} 0100 \\ + 1000 \\ \hline 1100 \\ + 0110 \\ \hline 0001 \ 0010 \end{array} \quad \begin{array}{l} \text{Binary Sum} > 9 \\ \text{Add 6} \\ \text{Ans. BCD sum} \end{array}$$

1 2

③ $8 + 9$

Solution

$$\begin{array}{r} 8 \\ + 9 \\ \hline 17 \end{array} \quad \begin{array}{r} 1000 \\ + 1001 \\ \hline 10001 \\ + 0110 \\ \hline 0001 \ 0111 \end{array} \quad \begin{array}{l} \text{Binary sum} < 9, \text{ with carry} \\ \text{Add 6} \\ \text{Ans BCD sum} \end{array}$$

1 7

④ Using BCD Addition, add $24 + 18$

Solution:

$$\begin{array}{r} 24 \\ + 18 \\ \hline 42 \end{array} \quad \begin{array}{r} 0010 \\ + 0001 \\ \hline 0100 \\ + 0100 \\ \hline 0100 \end{array} \quad \begin{array}{r} 0100 \\ 1000 \\ \hline 1100 \\ 0110 \\ \hline 0010 \end{array} \quad \begin{array}{l} \text{Binary sum} > 9 \\ \text{Add 6} \\ \text{Ans BCD sum} \end{array}$$

4 2

⑤ Using BCD, add $184 + 576$

Solution:

$$\begin{array}{r} 184 \\ + 576 \\ \hline 760 \end{array} \quad \begin{array}{r} 0001 \\ + 0101 \\ \hline 0111 \\ + 0110 \\ \hline 0111 \end{array} \quad \begin{array}{r} 1000 \\ 0111 \\ \hline 10000 \\ 0110 \\ \hline 0110 \end{array} \quad \begin{array}{r} 0100 \\ 0110 \\ \hline 1010 \\ 0110 \\ \hline 0000 \end{array} \quad \begin{array}{l} \text{Sum} > 9 \\ \text{Add 6} \\ \text{Ans BCD sum} = 0111 \ 0110 \ 0000 \end{array}$$

7 6 0

⑥ Using BCD, add $48 + 58$

Solution:

$$\begin{array}{r} 48 \\ + 58 \\ \hline 106 \end{array} \quad \begin{array}{r} 0100 \\ + 0101 \\ \hline 1010 \\ + 0110 \\ \hline 0001 \ 0000 \end{array} \quad \begin{array}{r} 1000 \\ 1000 \\ \hline 10000 \\ 0110 \\ \hline 0110 \end{array} \quad \begin{array}{l} \text{Binary sum} < 9, \text{ with carry} \\ \text{Add 6} \\ \text{Ans BCD sum} \end{array}$$

1 0 6

⑦ Using BCD, add $175 + 326$

Solution:

$$\begin{array}{r} 175 \\ + 326 \\ \hline 501 \end{array} \quad \begin{array}{r} 0001 \\ + 0011 \\ \hline 0101 \\ + 0101 \\ \hline 0101 \end{array} \quad \begin{array}{r} 0111 \\ 0010 \\ \hline 1010 \\ 0110 \\ \hline 0000 \end{array} \quad \begin{array}{r} 0101 \\ 0110 \\ \hline 1011 \\ 0110 \\ \hline 0001 \end{array} \quad \begin{array}{l} \text{Binary sum} > 9 \\ \text{Add 6} \\ \text{Ans BCD sum} \end{array}$$

5 0 1

⑧ Using BCD, add $589 + 199$

Solution:

$$\begin{array}{r} 589 \\ + 199 \\ \hline 788 \end{array} \quad \begin{array}{r} 0101 \\ + 0001 \\ \hline 0111 \\ + 0111 \\ \hline 0111 \end{array} \quad \begin{array}{r} 1000 \\ 1001 \\ \hline 10010 \\ 0110 \\ \hline 1000 \end{array} \quad \begin{array}{r} 1001 \\ 1001 \\ \hline 10010 \\ 0610 \\ \hline 1000 \end{array} \quad \begin{array}{l} \text{Sum} < 9, \text{ carry} \\ \text{Add 6} \\ \text{Ans BCD sum} \end{array}$$

7 8 8